

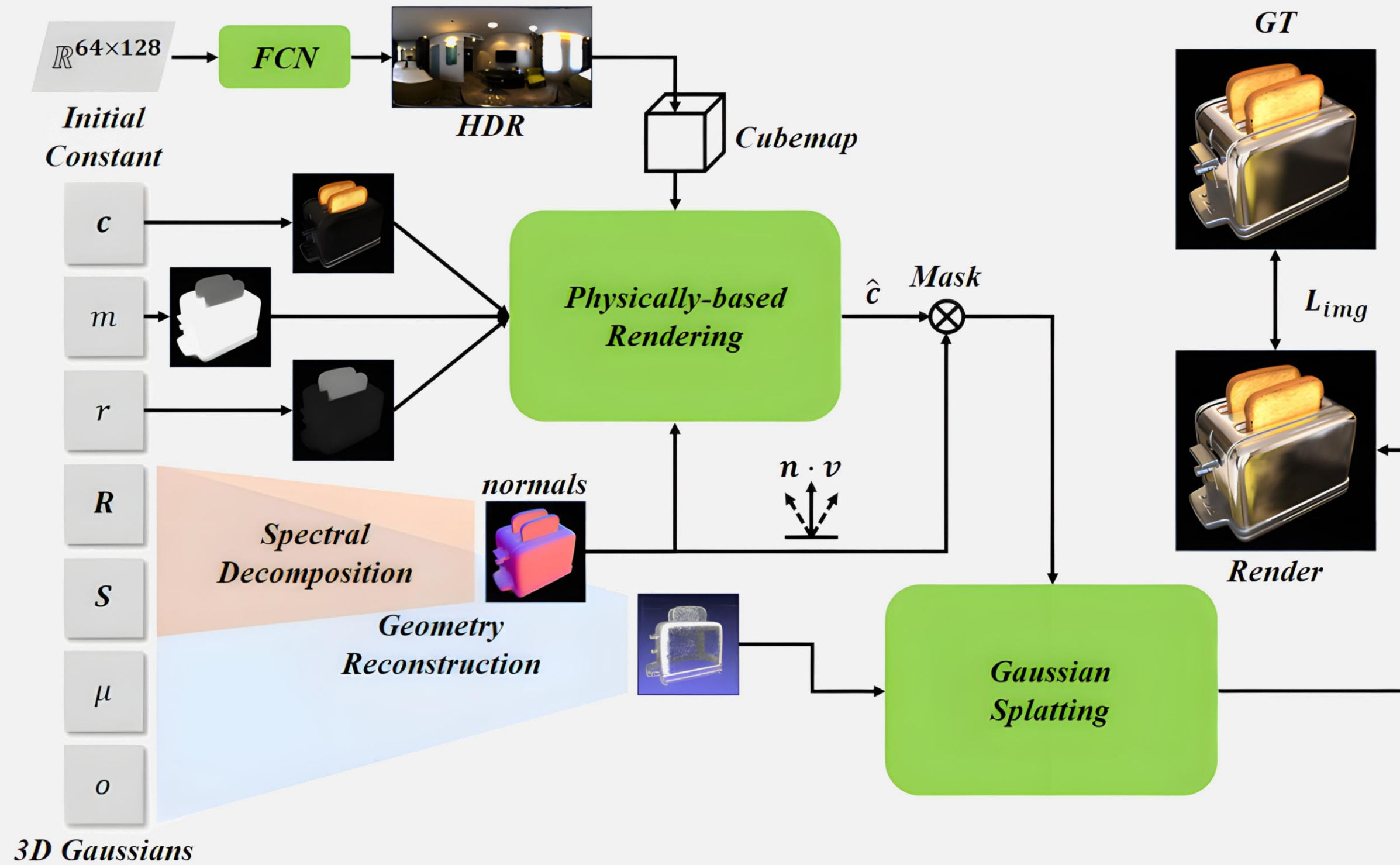
Motivation

Standard 3DGS **entangles lighting and materials**, preventing realistic editing and relighting.

Contribution

We leverage **diffusion priors** to decouple baked-in lighting into explicit PBR parameters for **accurate scene relighting**.

Gaussian Inverse Rendering (GIR)



Limitation: Albedo-illumination ambiguity, approximated light transport

Prior Loss Functions

$$\mathcal{L}_{\text{albedo}} = 1 - \text{ZNCC}(a, \hat{a})$$

$$\mathcal{L}_{\text{normal}} = 1 - (n_i \cdot \hat{n}_i) / (\|n_i\| \|\hat{n}_i\|)$$

$$\mathcal{L}_{\text{metallic}} = \text{Huber}(m, \hat{m})$$

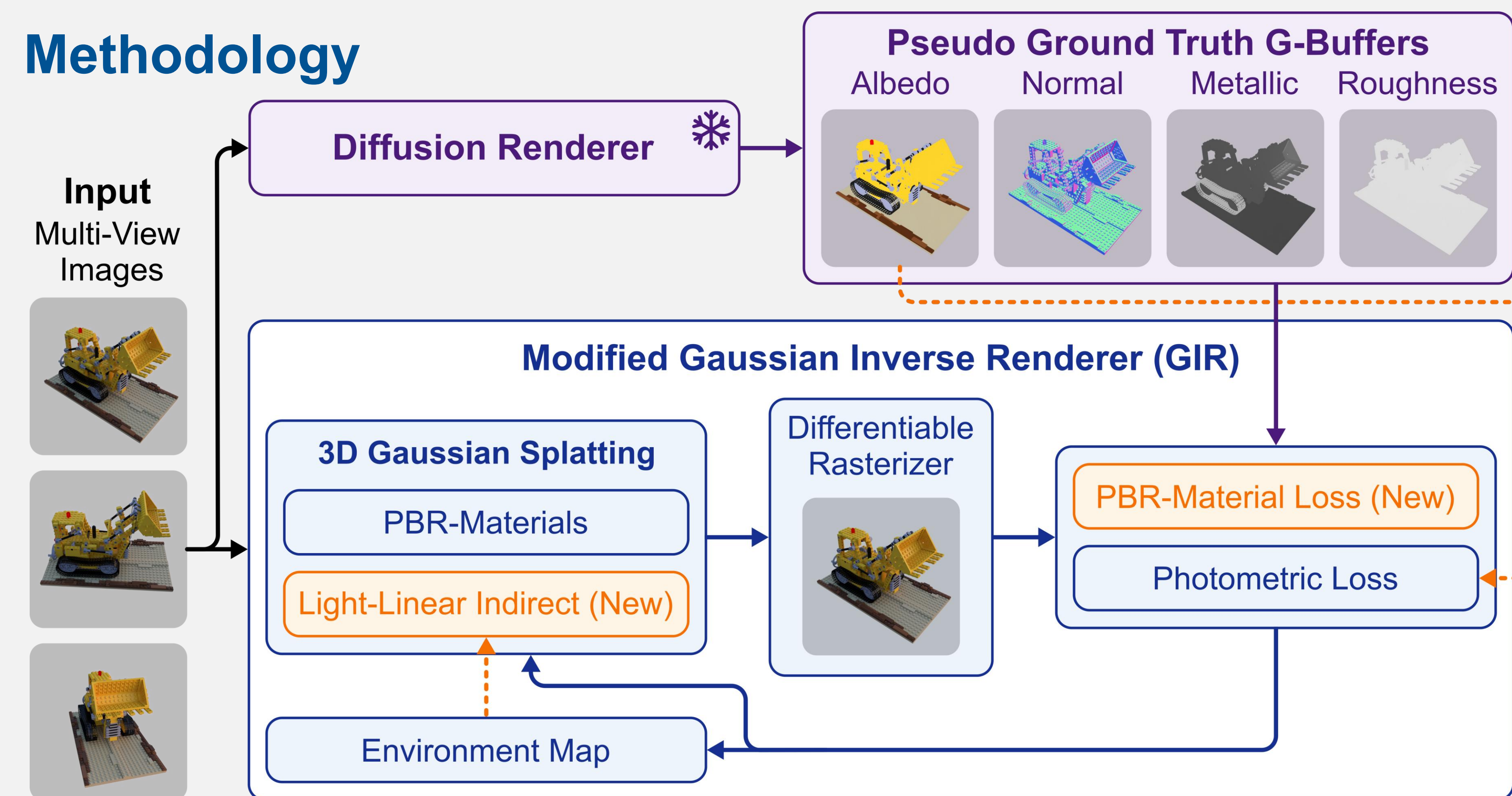
$$\mathcal{L}_{\text{roughness}} = \text{Huber}(r, \hat{r})$$

$$\mathcal{L} = \mathcal{L}_{\text{photo}} + \sum_{x \in \{a, n, m, r\}} \lambda_x \mathcal{L}_x$$

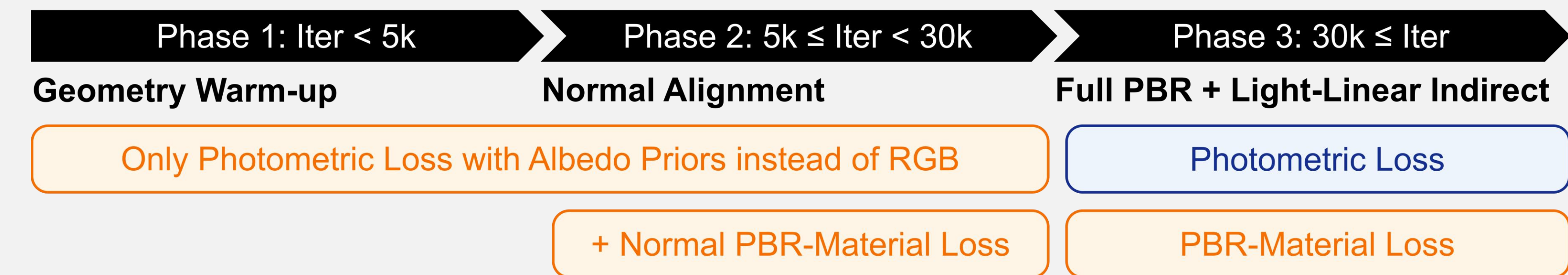
References:

- [1] Shi, Yahao, et al. "Gir: 3d gaussian inverse rendering for relightable scene factorization." *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
 [2] Liang, Ruofan, et al. "Diffusion renderer: Neural inverse and forward rendering with video diffusion models." *Proceedings of the Computer Vision and Pattern Recognition Conference*. 2025.

Methodology



Modified Training Process

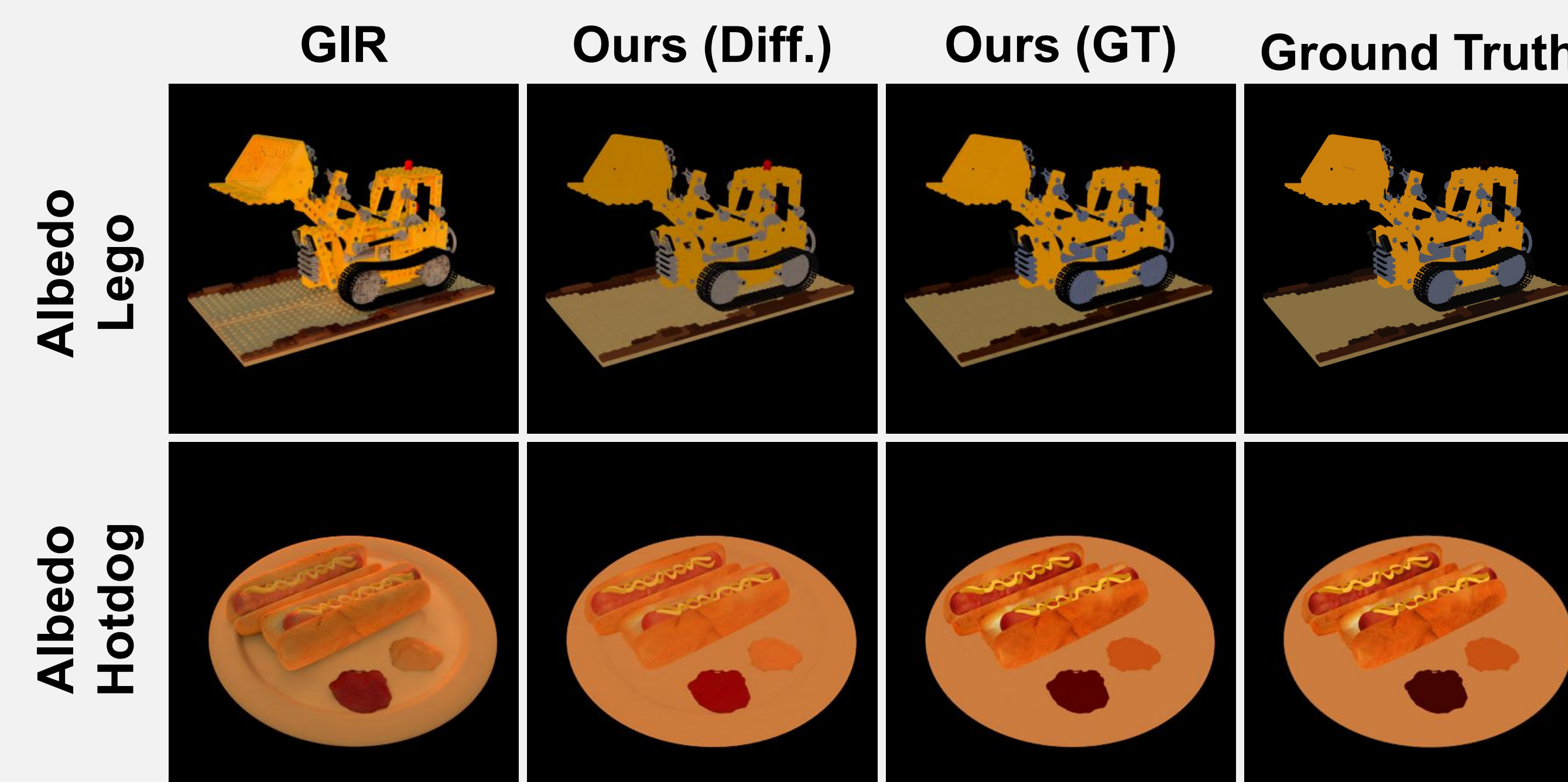


Quantitative Results

Method	Mean Albedo			Mean Normal		Mean Relighting		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MAE $\downarrow$		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
GIR	25.80	0.91	0.10	6.88		26.221	0.892	0.081
Ours (Diffusion priors)	30.02	0.95	0.04	4.63		27.718	0.913	0.075
Ours (GT priors)	36.92	0.97	0.02	1.89		27.277	0.912	0.075

Results: Material Decomposition

Albedo images are scale-aligned to the ground truth.



Relighting

